IT & FULL STACK DEVELOPMENT

Build Digital Skills. Develop Real Applications. Prepare for a Technology Career.

Aayush Institute of Professional Studies offers career-oriented **IT & Full Stack Development training** designed for students, graduates, freshers, working professionals and learners looking to build practical skills for the technology industry.

The programme focuses on programming fundamentals, modern web technologies, frontend and backend development, databases, APIs, version control, testing, deployment and real-world project development.

Our learning approach combines **concepts, practical assignments, project work, industry-oriented exposure and career preparation**.

WHAT IS FULL STACK DEVELOPMENT?

Full Stack Development involves building complete web applications by working across both the **frontend and backend** of a digital product.

A Full Stack Developer typically works with:

Frontend
HTML5 → CSS3 → JavaScript → React / Other Frontend Technologies

Backend
Node.js / Python / Java / PHP → Server-Side Development → Business Logic

Database
MySQL / PostgreSQL / MongoDB and other database technologies

Application Layer
REST APIs → Authentication → Integration → Testing

Development & Deployment
Git & GitHub → Debugging → Hosting → Deployment

This enables learners to understand how a complete web application is designed, developed, tested and deployed.

WHO CAN LEARN?

The programme can be suitable for:

- 12th Pass Students
- ITI Students
- Polytechnic / Diploma Students
- BCA Students
- B.Sc. IT / Computer Science Students
- B.Tech / B.E. Students
- MCA Students
- M.Tech Students
- Graduates from Relevant Backgrounds
- Working Professionals
- Freshers
- Career Changers interested in Technology

Prior programming experience may be helpful for advanced modules, but foundation-level learning can be started by eligible beginners.

COMPLETE LEARNING PATH

01. COMPUTER & PROGRAMMING FUNDAMENTALS

Build a strong foundation before moving into advanced development.

Topics Include:

- Computer Fundamentals
- Operating System Basics
- Internet & Web Fundamentals
- Software Development Concepts
- Programming Logic
- Algorithms & Problem Solving
- Basic Data Structures

- Development Environment
- Command Line Fundamentals
- Introduction to Git

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02. HTML5 — WEB STRUCTURE

Learn how modern web pages and application interfaces are structured.

Topics Include:

- HTML5 Fundamentals
- Semantic HTML
- Headings, Paragraphs & Lists
- Links & Navigation
- Images & Multimedia
- Tables
- Forms & Form Elements
- Page Structure
- Accessibility Basics
- Modern HTML Practices

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03. CSS3 — RESPONSIVE WEB DESIGN

Learn to create modern, responsive and user-friendly web interfaces.

Topics Include:

- CSS Fundamentals
- Selectors & Properties
- Box Model
- Typography

- Colors & Backgrounds
- Flexbox
- CSS Grid
- Responsive Design
- Media Queries
- Layout Techniques
- UI Styling
- Modern CSS Practices

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04. JAVASCRIPT

JavaScript forms the foundation of modern interactive web development.

Topics Include:

- JavaScript Fundamentals
- Variables & Data Types
- Operators
- Conditions & Loops
- Functions
- Arrays & Objects
- DOM Manipulation
- Events
- Forms & Validation
- ES6+ Concepts
- Asynchronous JavaScript
- Promises
- API Integration

- Error Handling

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05. FRONTEND DEVELOPMENT

Move from basic web pages to application-level frontend development.

React JS Topics May Include:

- React Fundamentals
- Components
- JSX
- Props & State
- Hooks
- Forms
- Event Handling
- Routing
- API Integration
- State Management Fundamentals
- Reusable Components
- Frontend Project Development

Additional Technologies

Depending on the selected programme:

- TypeScript
- Modern JavaScript
- UI Component Libraries
- CSS Frameworks
- Git & GitHub
- Frontend Development Tools

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06. BACKEND DEVELOPMENT

Learn how applications process data, manage users and communicate with databases.

Node.js / Express Topics:

- Node.js Fundamentals
- Express.js
- Server-Side Development
- Routing
- Middleware
- Request & Response
- REST API Development
- Authentication
- Authorization
- Error Handling
- Backend Project Structure

Alternative Backend Tracks

Depending on the selected programme, learners may work with:

Python Full Stack
Python + Django / Flask + Database + REST APIs

Java Full Stack
Java + Spring / Spring Boot + Database + REST APIs

PHP Full Stack
PHP + MySQL + Web Application Development

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07. DATABASE MANAGEMENT

Understand how application data is stored, managed and retrieved.

SQL Technologies

- MySQL
- PostgreSQL
- SQL Fundamentals
- Database Design
- Tables & Relationships
- CRUD Operations
- Queries
- Joins
- Data Management

NoSQL

- MongoDB Fundamentals
- Collections & Documents
- CRUD Operations
- Application Integration

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08. API DEVELOPMENT & INTEGRATION

Learn how different applications and services communicate with each other.

Topics Include:

- API Fundamentals
- REST Architecture
- HTTP Methods
- GET / POST / PUT / PATCH / DELETE
- JSON
- API Requests & Responses
- Authentication

- API Testing
- Third-Party API Integration
- Frontend–Backend Communication

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09. AUTHENTICATION & SECURITY FUNDAMENTALS

Understand the basic security concepts required for modern applications.

Topics May Include:

- User Registration & Login
- Authentication
- Authorization
- Password Security Fundamentals
- Sessions / Tokens
- API Security Basics
- Input Validation
- Common Web Security Risks
- Secure Development Practices

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10. GIT & GITHUB

Learn professional development practices used for managing source code.

Topics Include:

- Git Fundamentals
- Repository Management
- Commit & Branch
- Merge
- Pull Requests

- GitHub
- Code Collaboration
- Version Control
- Project Repository Management

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11. SOFTWARE DEVELOPMENT PRACTICES

Understand how a real software project moves from an idea to a working application.

Development Lifecycle:

Requirement → Planning → Design → Development → Testing → Deployment → Maintenance

Learners are introduced to practical development practices including:

- Requirement Understanding
- Project Planning
- Folder & Code Structure
- Version Control
- Debugging
- Testing
- Documentation
- Deployment Preparation

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12. TESTING & DEBUGGING

Learn how to identify and resolve problems in applications.

Topics Include:

- Debugging Fundamentals
- Browser Developer Tools
- Error Identification

- Code Testing
- Functional Testing Basics
- API Testing
- Bug Fixing
- Performance Fundamentals

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13. DEPLOYMENT & HOSTING FUNDAMENTALS

Learn the basic process of making applications available online.

Topics May Include:

- Hosting Fundamentals
- Domain & Hosting Concepts
- Web Server Basics
- Environment Configuration
- Deployment Workflow
- Frontend Deployment
- Backend Deployment
- Database Connectivity
- Basic Cloud Deployment Concepts

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14. REAL PROJECT DEVELOPMENT

Practical project work is an important part of the learning process.

Projects may be developed according to the learner's level and selected technology track.

Beginner Projects

- Personal Portfolio Website
- Business Website

- Landing Page
- Responsive Website
- Contact / Enquiry Application

Intermediate Projects

- Blog Application
- Student Management System
- Inventory Management System
- E-Commerce Website
- Appointment Booking System

Advanced Projects

- Full Stack E-Commerce Application
- CRM Application
- Business Management System
- Online Learning Platform
- Service Booking Platform
- API-Based Web Application

Project availability and scope may vary according to the selected course and training programme.

TECHNOLOGY TRACKS

MERN STACK DEVELOPMENT

MongoDB + Express.js + React + Node.js

Suitable for learners interested in modern JavaScript-based full stack development.

PYTHON FULL STACK DEVELOPMENT

HTML + CSS + JavaScript + Python + Django/Flask + Database + REST API

Suitable for learners interested in Python-based application development.

JAVA FULL STACK DEVELOPMENT

HTML + CSS + JavaScript + Java + Spring/Spring Boot + Database + REST API

Suitable for learners looking to build a foundation in Java-based enterprise and web application development.

PHP FULL STACK DEVELOPMENT

HTML + CSS + JavaScript + PHP + MySQL + Web Application Development

Suitable for learners interested in PHP-based web development.

SKILLS YOU CAN DEVELOP

Depending on the selected programme, learners can develop skills in:

- Computer & IT Fundamentals
- Programming
- Web Development
- Frontend Development
- Backend Development
- Full Stack Development
- Database Management
- API Development
- Git & GitHub
- Debugging
- Testing
- Deployment
- Problem Solving

- Software Development Practices
- Project Development

CAREER PATHWAYS

Full Stack and IT development skills can support entry-level career pathways such as:

- Web Developer
- Frontend Developer
- Backend Developer
- Full Stack Developer
- React Developer
- Python Developer
- Java Developer
- PHP Developer
- Software Developer
- Junior Software Engineer
- Application Developer
- API Developer
- Web Application Developer

Actual job titles, eligibility and employer requirements vary by organisation, role and candidate profile.

LIVE PROJECT & PRACTICAL EXPOSURE

Where included in the selected programme, learners may receive opportunities for:

- Live Project Training
- Practical Assignments
- Industry-Oriented Projects

- Mentor-Guided Development
- Technical Documentation
- Testing & Debugging
- Project Presentation
- Portfolio Development
- Exposure to Professional Development Tools

The objective is to help learners move beyond classroom concepts and understand how technology is applied in practical projects.

CAREER GUIDANCE & PLACEMENT SUPPORT

Aayush Institute provides **career guidance at no additional cost** to help learners understand suitable technology pathways based on their qualification, interests, skills and career objectives.

Career support may include:

- Career Assessment
- Course & Career Guidance
- Resume / CV Preparation
- Portfolio Guidance
- Technical Interview Preparation
- HR Interview Preparation
- Mock Interviews
- Communication & Workplace Skills
- Job Role Guidance
- Recruitment Opportunity Updates
- Placement Assistance

Placement Information

Placement assistance and employer opportunities may be provided based on programme, eligibility and available opportunities.

Placement is not guaranteed. Selection depends on the candidate's skills, qualification, performance, interview, employer requirements, vacancies and other selection criteria.

INTERNSHIP OPPORTUNITIES

Eligible learners may also receive information about internship opportunities where available.

Internships may include:

- Technical Internships
- Software Development Internships
- Web Development Internships
- Data / Technology Internships
- Live Project Internships

Paid internships may be available with certain organisations, subject to eligibility, availability and the company's selection process.

Any stipend or compensation is determined by the organisation offering the internship.

COURSE LEVELS

FOUNDATION LEVEL

For beginners looking to develop basic computer, programming and web development skills.

PROFESSIONAL LEVEL

For learners who want structured training in frontend, backend and application development.

ADVANCED FULL STACK LEVEL

For learners looking to work across frontend, backend, database, APIs, Git and deployment.

SPECIALISED TECHNOLOGY TRACKS

Focused learning pathways such as:

- MERN Stack
- Python Full Stack

- Java Full Stack
- PHP Full Stack
- React Development
- Backend Development
- Web Development

WHO SHOULD CHOOSE IT?

IT & Full Stack Development can be considered by learners who:

- Want to enter the IT industry
- Are interested in software development
- Want to learn web technologies
- Want to build websites and applications
- Want practical programming skills
- Want to develop a technology portfolio
- Want to upgrade existing IT skills
- Are planning a career transition into technology

YOUR LEARNING JOURNEY

BEGINNER

Understand Computers & Programming

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PROGRAMMER

Build Programming Logic

↓

WEB DEVELOPER

Learn HTML, CSS & JavaScript

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APPLICATION DEVELOPER

Build Frontend & Backend Applications

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FULL STACK DEVELOPER

Work with Frontend + Backend + Database + APIs

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PROJECT-READY PROFESSIONAL

Develop Projects, Build Portfolio & Prepare for Career Opportunities

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WHY LEARN WITH AAYUSH?

ESTABLISHED SINCE 2002

More than two decades of association with education and professional skill development.

PRACTICAL LEARNING

Training designed around concepts, assignments and application of skills.

INDUSTRY-ORIENTED TECHNOLOGIES

Learning pathways aligned with commonly used development technologies and professional practices.

LIVE PROJECT EXPOSURE

Practical project opportunities may be available depending on the selected programme.

CAREER GUIDANCE

Career guidance is provided at no additional cost.

INTERNSHIP OPPORTUNITIES

Eligible learners may receive access to available internship opportunities, subject to organisation requirements and selection.

PLACEMENT SUPPORT

Career preparation and placement assistance are provided, subject to eligibility and available employer opportunities.

BUILD YOUR TECHNOLOGY CAREER

Learn the fundamentals.
Develop practical skills.
Build real projects.
Create your portfolio.
Prepare for technology opportunities.

Start Your IT & Full Stack Development Journey with Aayush Institute of Professional Studies.

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